

ambiguous grammar - Results



Attempt 1 of 1

Written Jan 29, 2026 10:37 AM - Jan 29, 2026 11:10 AM

Attempt Score **8 / 11 - 72.73 %**

Overall Grade (Highest Attempt) **8 / 11 - 72.73 %**

Question 1

0 / 1 point

How many parse trees does $()()$ have for the grammar

$S \rightarrow SS \mid (S) \mid \epsilon$?

- ☐ 1
-  ☒ 2
- ☐ 3
- ☐ 0
-  ☐ infinite

Question 2

0 / 1 point

How many parse trees does $()()$ have for the grammar

$S \rightarrow (S)S \mid \epsilon$?

-  ☐ 1
-  ☒ 2

- ☐ 3
- ☐ 0
- ☐ infinite

Question 3

1 / 1 point

. Which grammar is ambiguous?

- ☐ $S \rightarrow aSb \mid \epsilon$
- ☐ $S \rightarrow aS \mid b$
- ☐ $S \rightarrow abS \mid \epsilon$
- ✓ ☒ $S \rightarrow SS \mid a$

Question 4

0 / 1 point

Which string demonstrates ambiguity in the grammar below?

$S \rightarrow SS \mid a$

- ☐ a
- ☐ aa
- ➡ ☐ aaa
- ✗ ☒ all above

Question 5

1 / 1 point

Which property is NOT sufficient to prove ambiguity?

- ☐ Two distinct parse trees
- ☐ Two different leftmost derivations

- ☐ Two different rightmost derivations
- ✓ ☒ Two different derivations
- ☐ None of the above

Question 6

1 / 1 point

If a grammar is ambiguous, then:

- ☐ All equivalent grammars are ambiguous
- ☐ No equivalent unambiguous grammar exists
- ✓ ☒ There may exist an equivalent unambiguous grammar
- ☐ The language itself is ambiguous

Question 7

1 / 1 point

Give the corresponding regular grammar for

$S \rightarrow SS \mid a$

- ✓ ☒ $S \rightarrow aS \mid a$
- ☐ $S \rightarrow aS \mid \epsilon$
- ☐ $S \rightarrow aaS \mid a$
- ☐ $S \rightarrow aaS \mid \epsilon$

Question 8

1 / 1 point

Which of the following CFGs can be transformed into a regular grammar?

- ☐ $S \rightarrow aSb \mid \epsilon$
- ✓ ☒ $S \rightarrow aaS \mid aa$
- ☐ $S \rightarrow aSa \mid bSb \mid a \mid b$
- ☐ $S \rightarrow SaSb \mid a$

Question 9

1 / 1 point

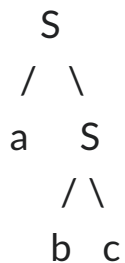
Given a parse tree of string s . s can be obtained from the tree by

- ☐ Preorder traversal
- ☐ Postorder traversal
- ☐ Inorder traversal
- ✓ ☒ concatenating leaf nodes left to right

Question 10

1 / 1 point

Given the following parse tree:



Which is the yield of the parse tree?

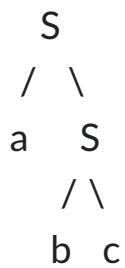
- ✓ ☒ abc
- ☐ aSSbc

- ☐ SaSbc
- ☐ aSbSc

Question 11

1 / 1 point

Given the following parse tree:



Which may not be in the corresponding grammar?

- ☐ $S \rightarrow aS$
- ☐ $S \rightarrow bc$
- ☒ $S \rightarrow aSbc$
- ☐ more than one answer

Done